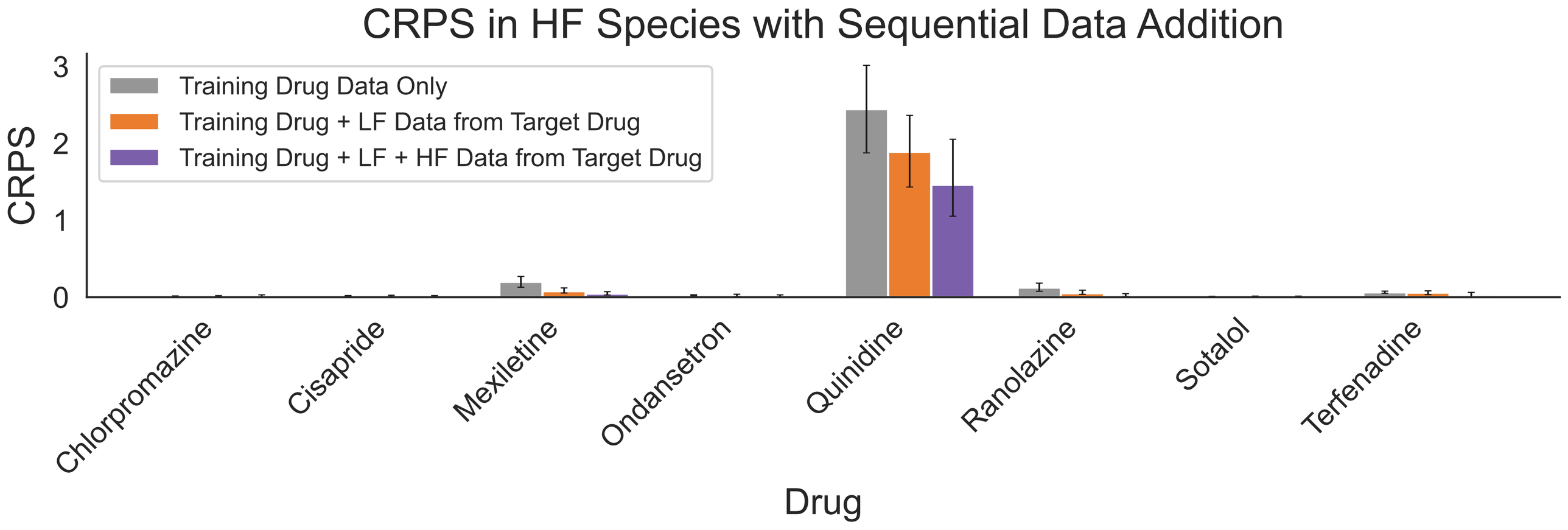
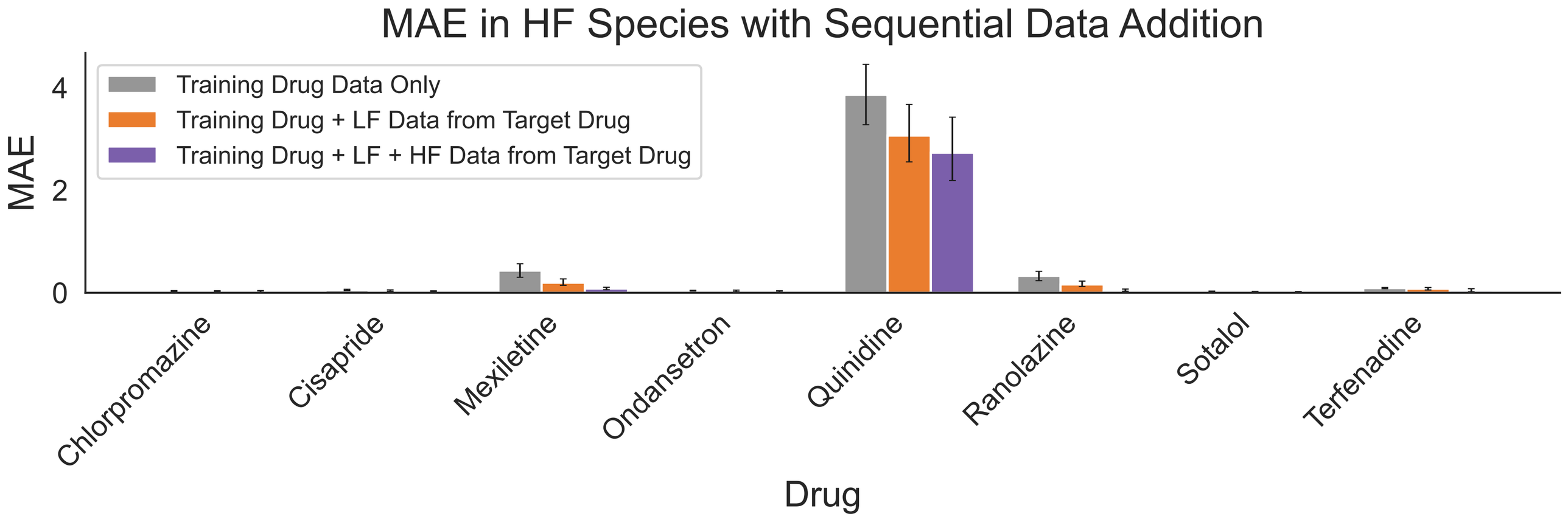
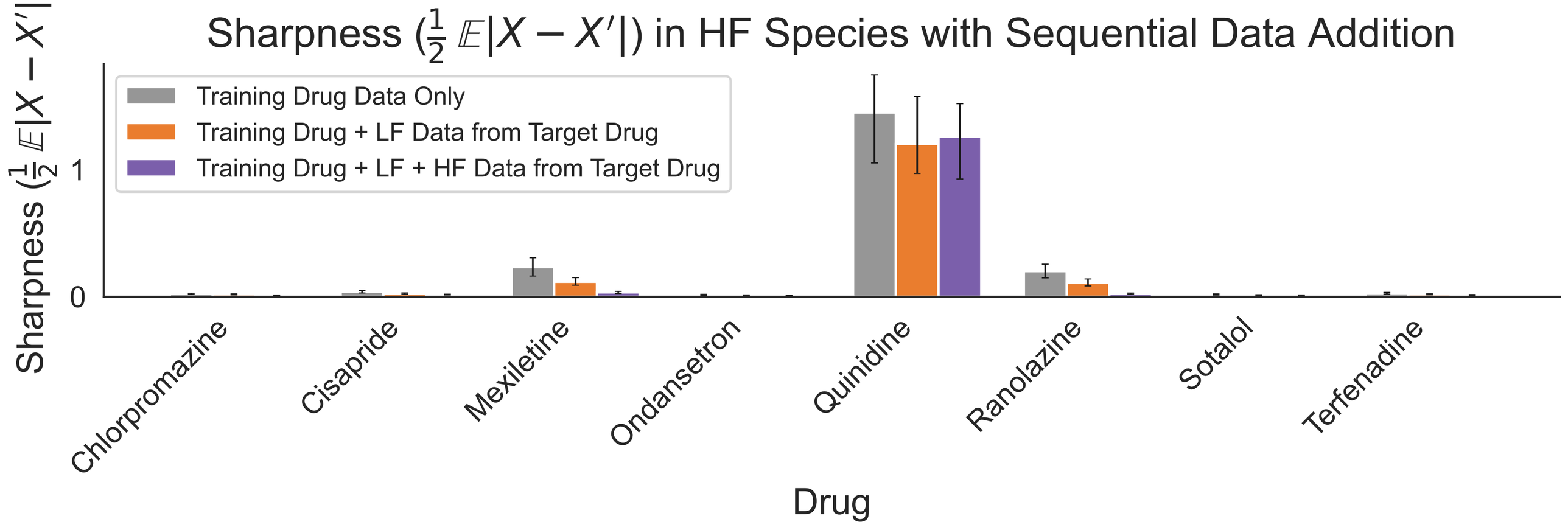
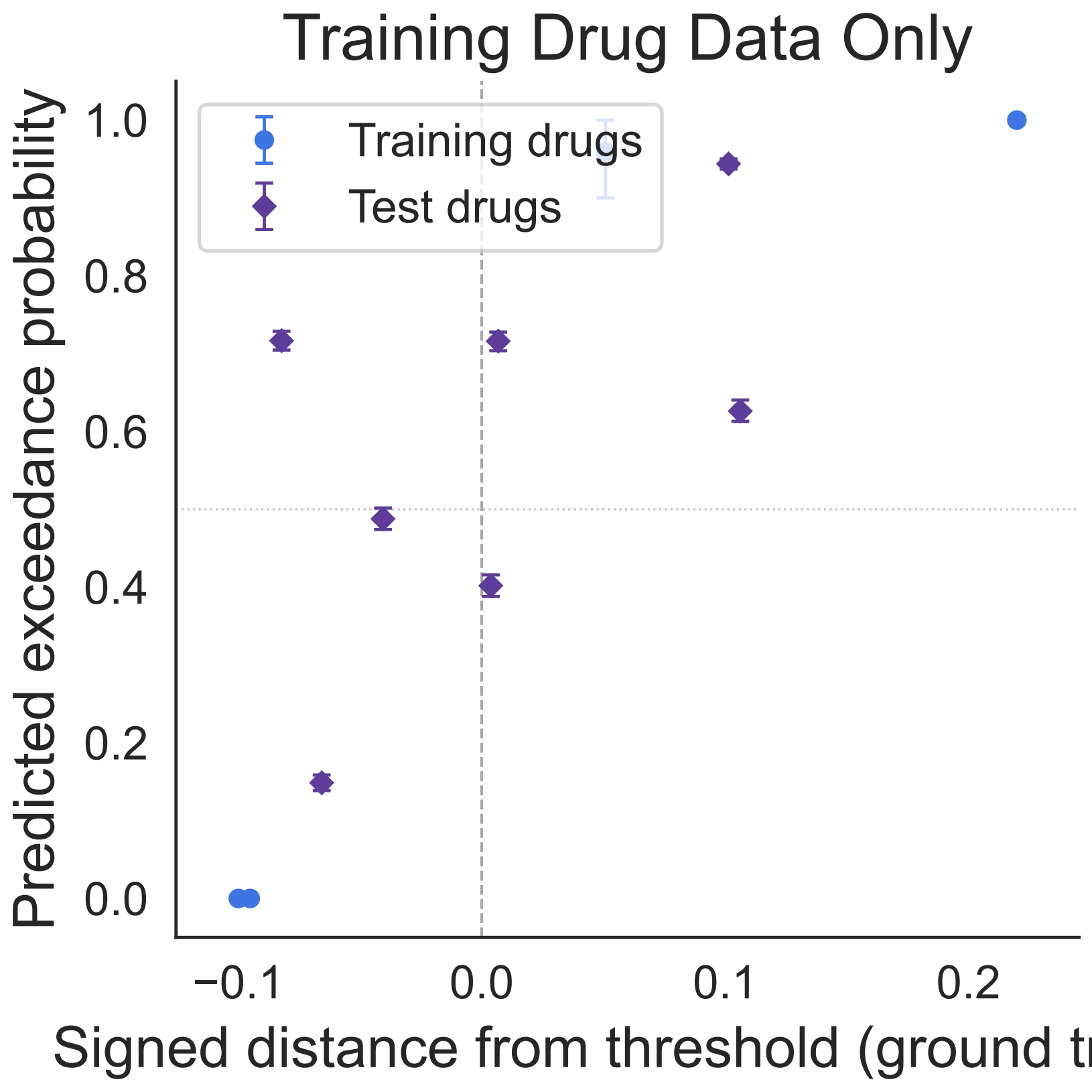

CRPS in HF Species with Sequential Data Addition
Training Drug Data Only
Training Drug + LF Data from Target Drug
Training Drug + LF + HF Data from Target Drug
CRPS
0
1
2
3
Chlorpromazine
Cisapride
Mexiletine
Ondansetron
Quinidine
Ranolazine
Sotalol
Terfenadine
Drug

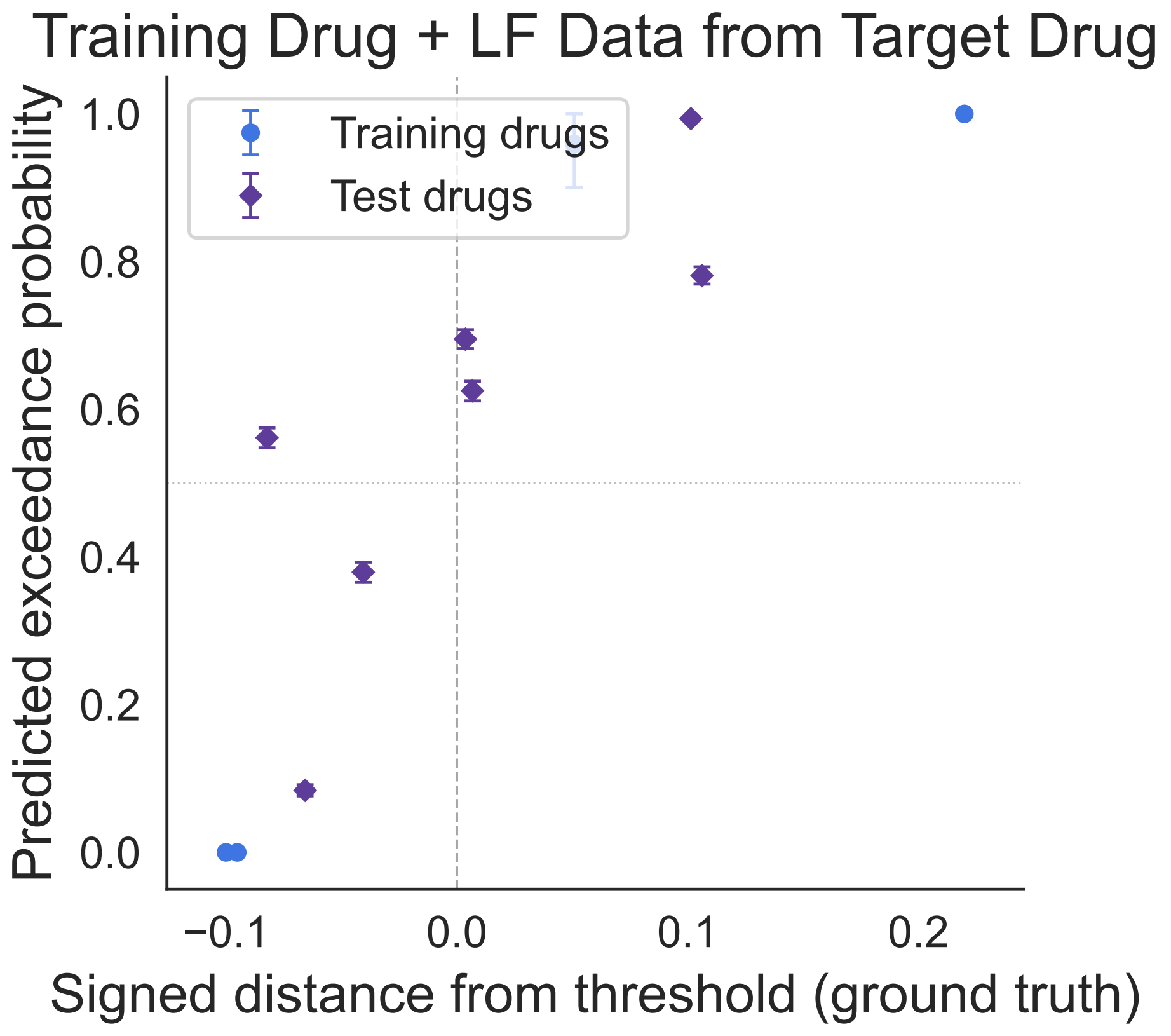

MAE in HF Species with Sequential Data Addition
Training Drug Data Only
Training Drug + LF Data from Target Drug
Training Drug + LF + HF Data from Target Drug
MAE
0
2
4
Chlorpromazine
Cisapride
Mexiletine
Ondansetron
Quinidine
Ranolazine
Sotalol
Terfenadine
Drug

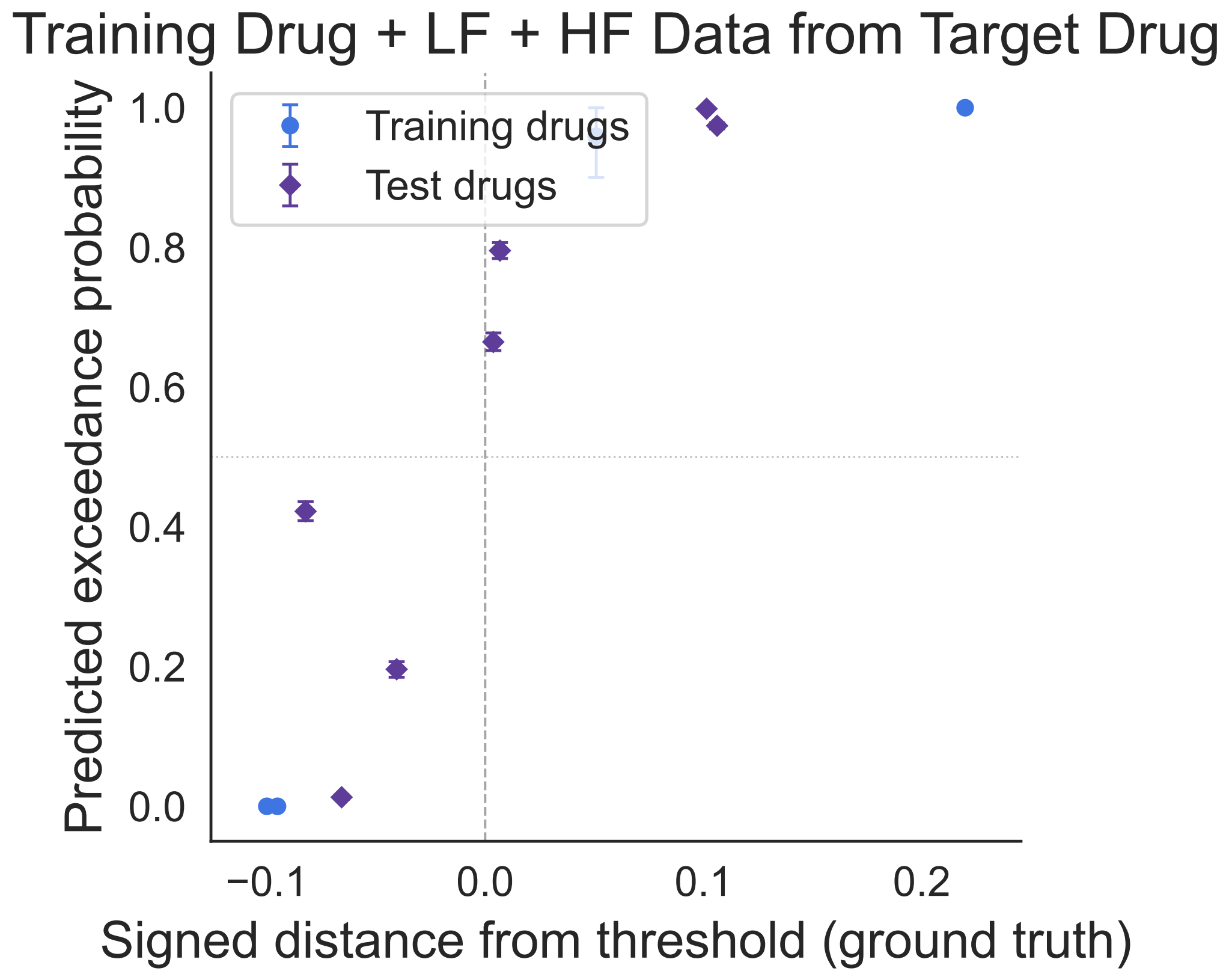

Sharpness ($\frac{1}{2}\,\mathbb{E}|X - X'|$) in HF Species with Sequential Data Addition
Training Drug Data Only
Training Drug + LF Data from Target Drug
Training Drug + LF + HF Data from Target Drug
Sharpness ($\frac{1}{2}\,\mathbb{E}|X - X'|$)
0
1
Chlorpromazine
Cisapride
Mexiletine
Ondansetron
Quinidine
Ranolazine
Sotalol
Terfenadine
Drug

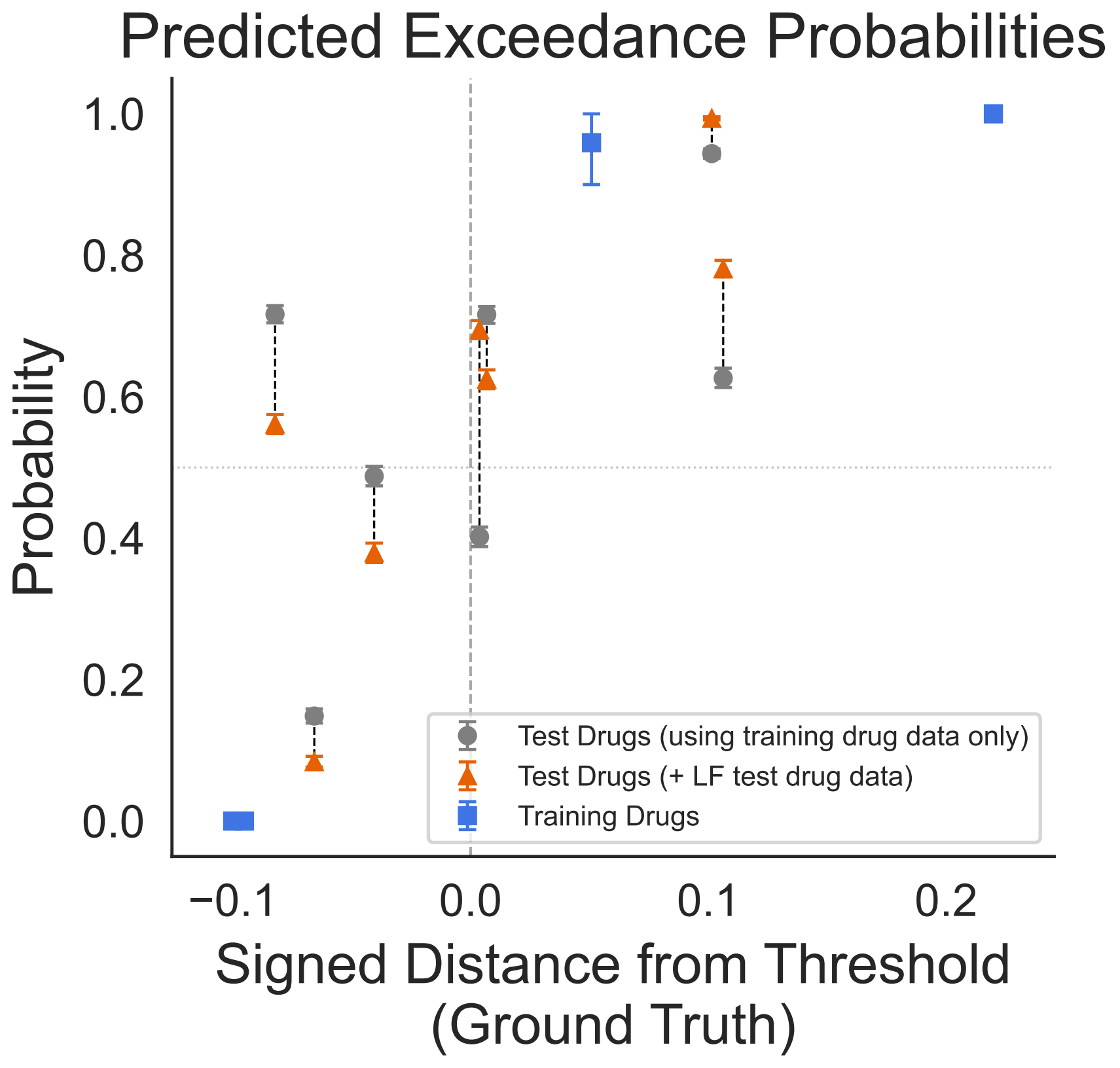
Exceedance Probability per Drug and Scenario (Median with 95% CI)

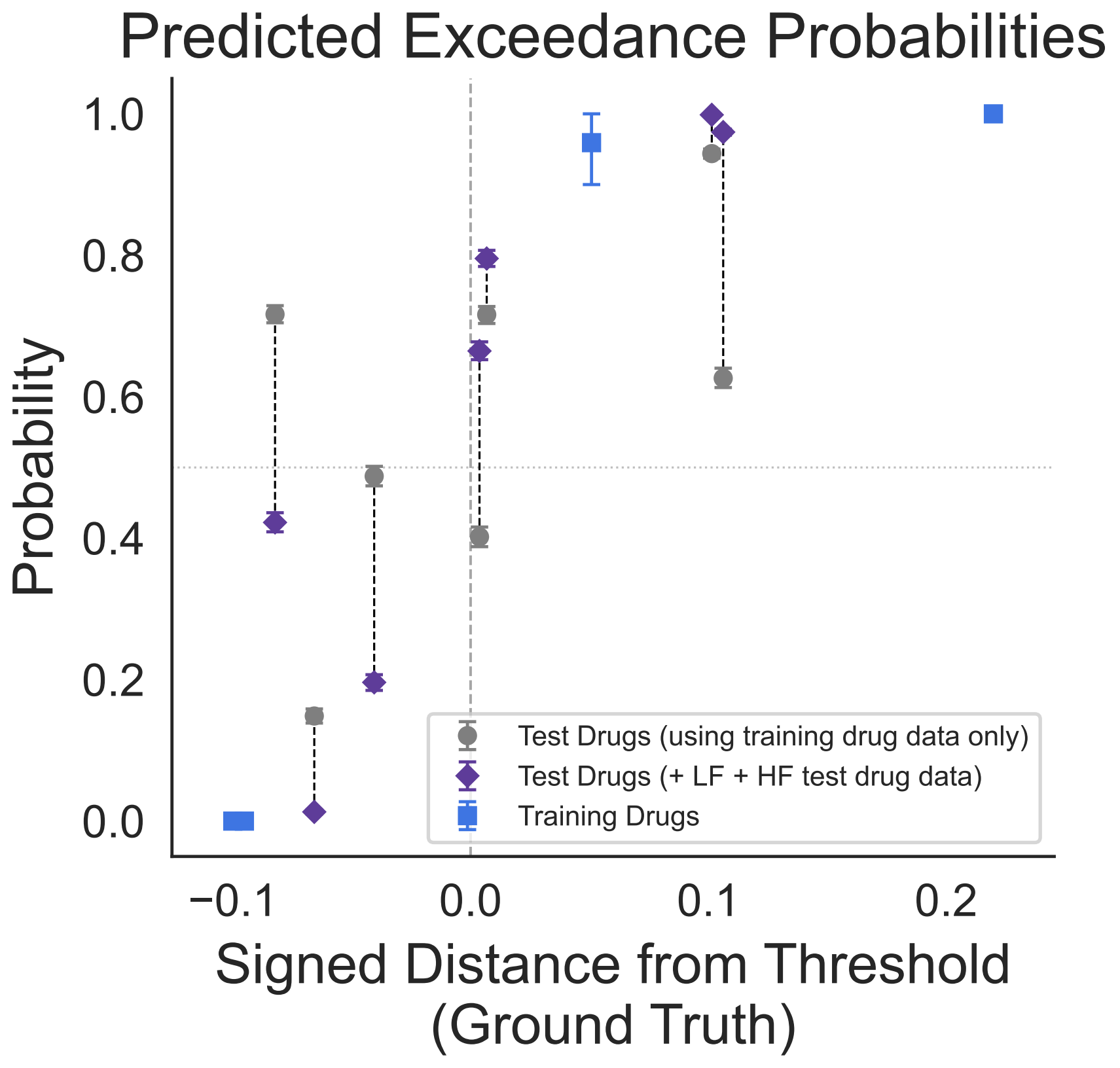
Drug	Training Drug Data Only	Training Drug + LF Data from Target Drug	Training Drug + LF + HF Data from Target Drug
Chlorpromazine	0.149 (0.139, 0.159)	0.084 (0.076, 0.092)	0.013 (0.010, 0.016)
Cisapride	0.944 (0.937, 0.950)	0.994 (0.991, 0.996)	0.999 (0.997, 1.000)
Mexiletine	0.717 (0.705, 0.729)	0.561 (0.548, 0.575)	0.422 (0.409, 0.436)
Ondansetron	0.488 (0.474, 0.502)	0.379 (0.366, 0.393)	0.196 (0.185, 0.207)
Quinidine	0.999 (0.997, 1.000)	1.000 (0.999, 1.000)	1.000 (1.000, 1.000)
Ranolazine	0.716 (0.704, 0.728)	0.625 (0.611, 0.638)	0.795 (0.784, 0.807)
Sotalol	0.402 (0.388, 0.416)	0.695 (0.682, 0.708)	0.665 (0.652, 0.678)
Terfenadine	0.626 (0.613, 0.640)	0.781 (0.770, 0.793)	0.974 (0.970, 0.978)

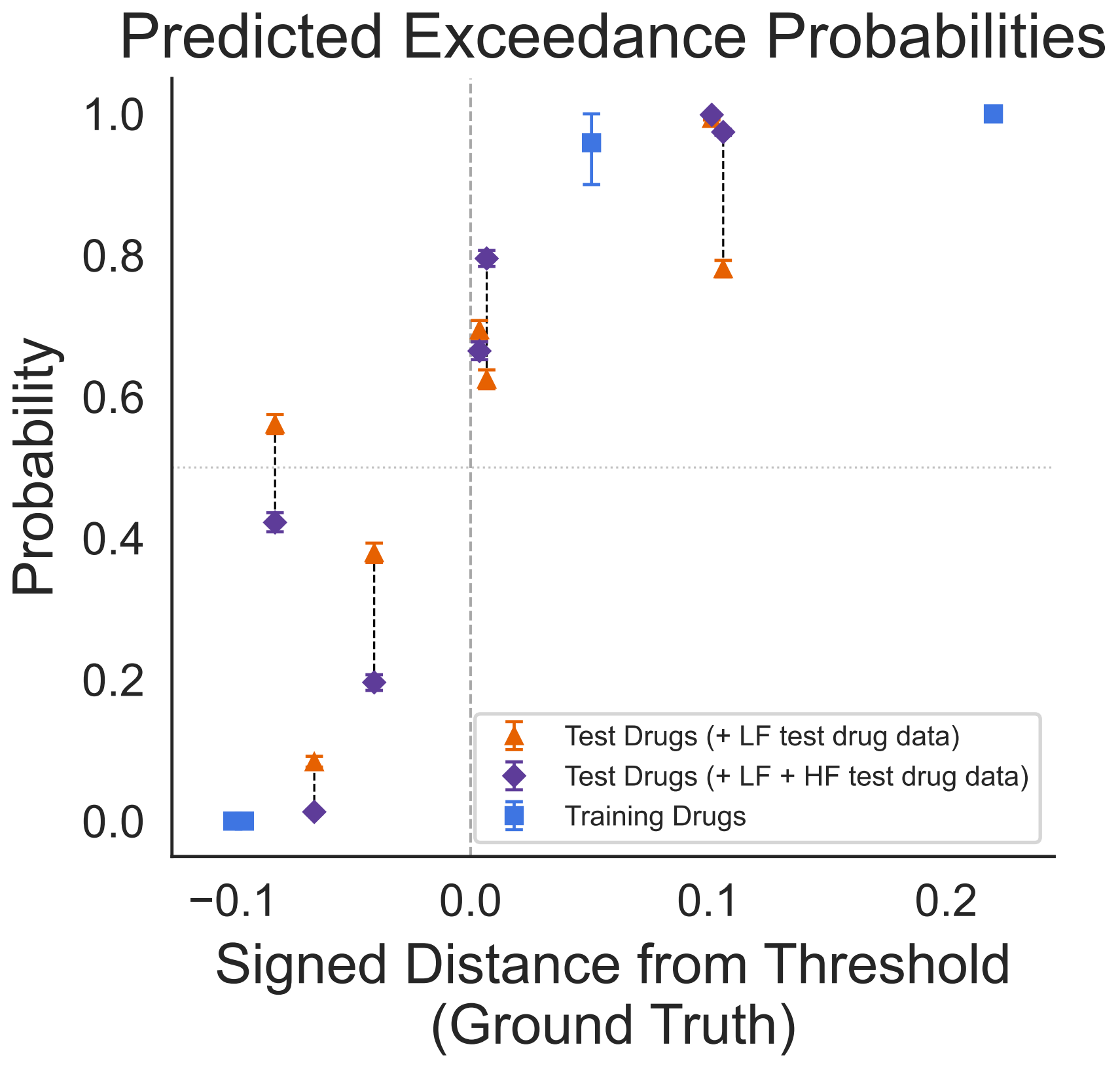

Training Drug Data Only
Training drugs
Test drugs
Predicted exceedance probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed distance from threshold (ground t

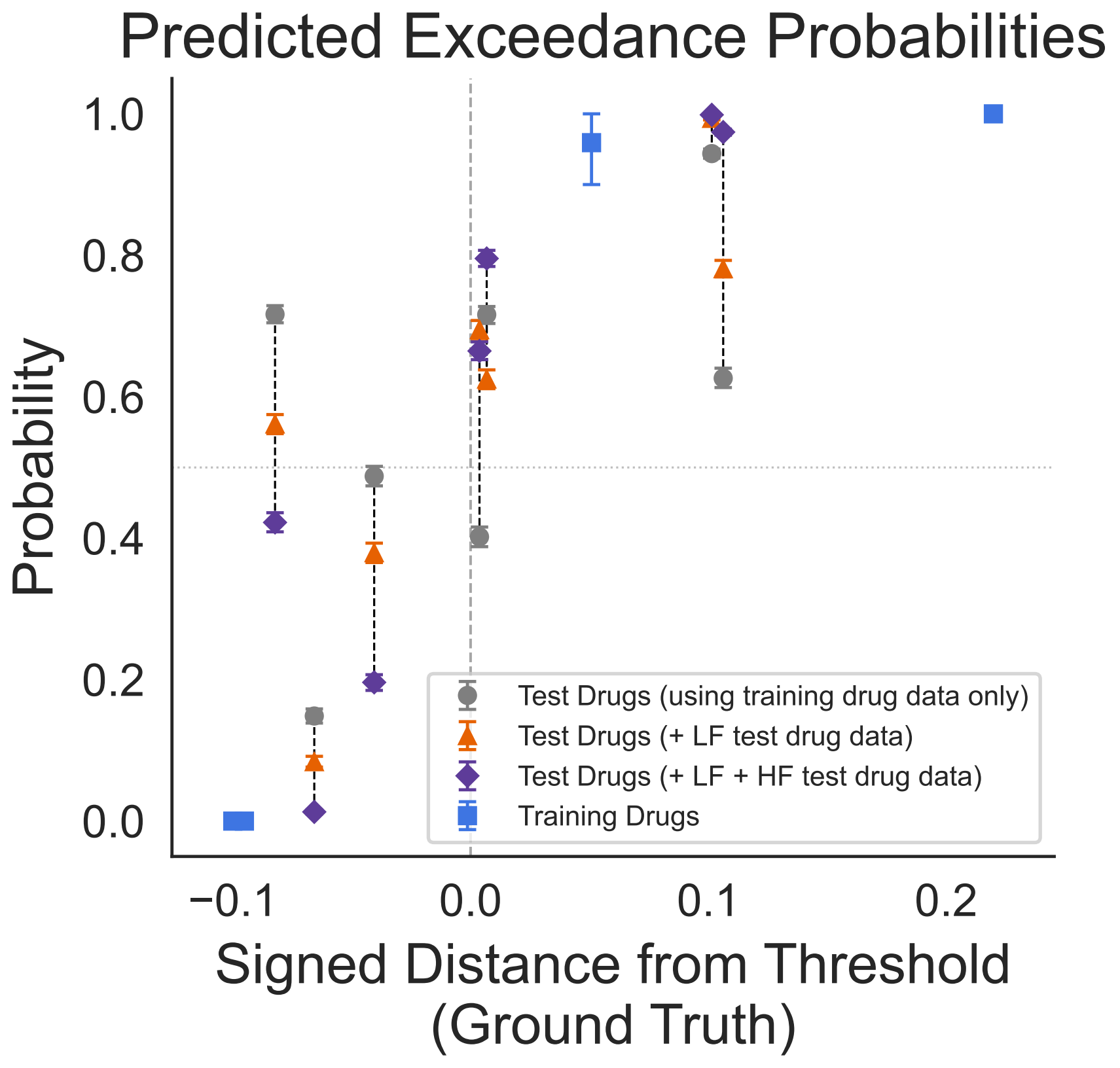

Training Drug + LF Data from Target Drug
Training drugs
Test drugs
Predicted exceedance probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed distance from threshold (ground truth)

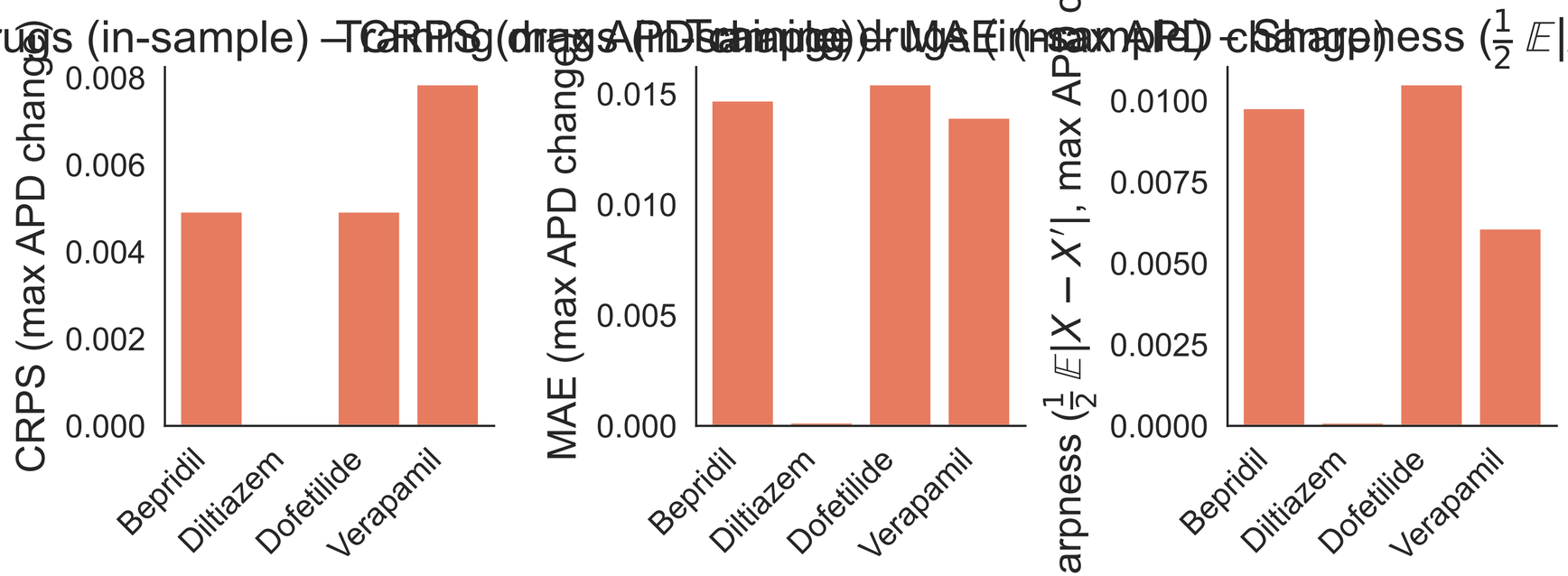

Training Drug + LF + HF Data from Target Drug
Predicted exceedance probability
1.0
0.8
0.6
0.4
0.2
0.0
Training drugs
Test drugs
−0.1
0.0
0.1
0.2
Signed distance from threshold (ground truth)


Predicted Exceedance Probabilities
Probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed Distance from Threshold
(Ground Truth)
Test Drugs (using training drug data only)
Test Drugs (+ LF test drug data)
Training Drugs


Predicted Exceedance Probabilities
Probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed Distance from Threshold
(Ground Truth)
Test Drugs (using training drug data only)
Test Drugs (+ LF + HF test drug data)
Training Drugs


Predicted Exceedance Probabilities
Probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed Distance from Threshold
(Ground Truth)
Test Drugs (+ LF test drug data)
Test Drugs (+ LF + HF test drug data)
Training Drugs


Predicted Exceedance Probabilities
Probability
1.0
0.8
0.6
0.4
0.2
0.0
−0.1
0.0
0.1
0.2
Signed Distance from Threshold
(Ground Truth)
Test Drugs (using training drug data only)
Test Drugs (+ LF test drug data)
Test Drugs (+ LF + HF test drug data)
Training Drugs


Training drugs (in-sample) – Tasking (drugs in-sample) – CRPS (max APD change)
Training drugs (in-sample) – Tasking (drugs in-sample) – MAE (max APD change)
Training drugs (in-sample) – Tasking (drugs in-sample) – Sharpness ($\frac{1}{2}\,\mathbb{E}|X - X'|$, max APD change)
CRPS (max APD change)
MAE (max APD change)
Sharpness ($\frac{1}{2}\,\mathbb{E}|X - X'|$, max APD change)
0.008
0.006
0.004
0.002
0.000
0.015
0.010
0.005
0.000
0.0100
0.0075
0.0050
0.0025
0.0000
Bepridil
Diltiazem
Dofetilide
Verapamil